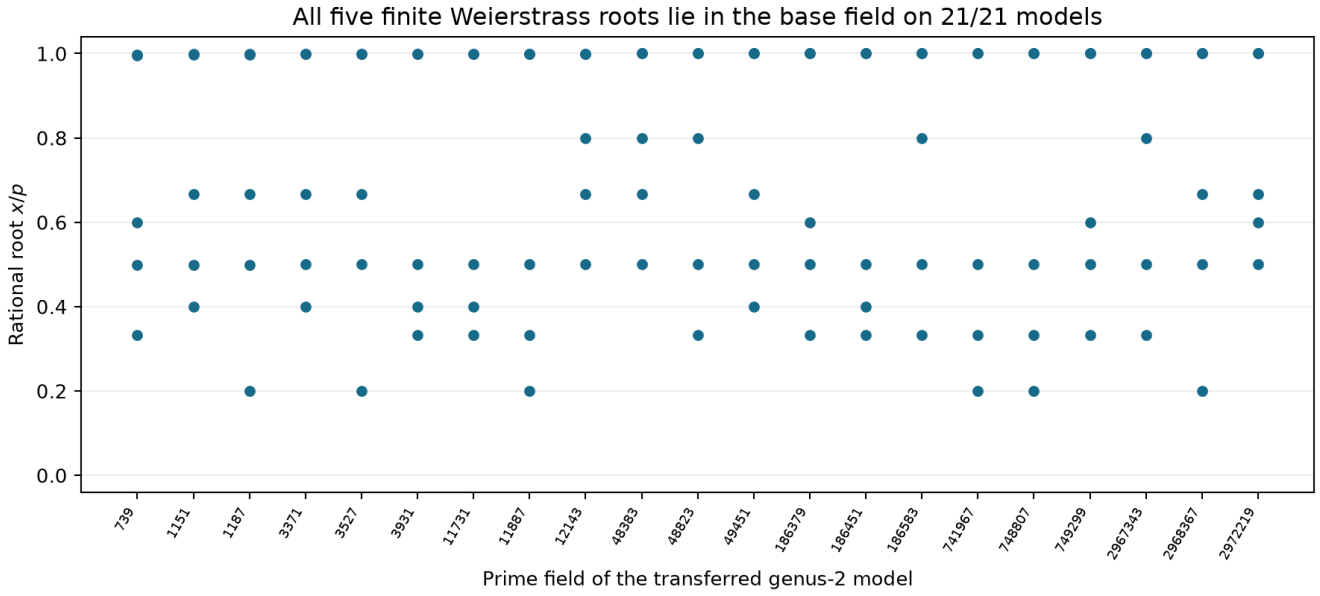
Fast-Kummer model applicability on 21 transferred curves

Exact arithmetic gate

21 October 2026

Scope. The exact genus-2 quintics from the frozen 21-model cohort were evaluated at every $x \in \mathbb{F}_p$. Every quintic had five distinct roots in $\mathbb{F}_p$. For each model, the program also checked the derivative at each root and reconstructed the original polynomial from the five linear factors. The native output was reproduced byte-for-byte in a separate run. Thus the six Weierstrass points, including infinity, are rational on all 21 models, yielding full rational two-torsion.

Exact gate	Passing models	Checked models
Five distinct finite rational roots	21	21
Polynomial reconstructed from roots	21	21
Unsigned ordinary-addition ambiguity witness	21	21



The rational two-torsion condition addresses the field of definition in Gaudry's fast-Kummer construction. The separate ambiguity witness shows why quotient points alone cannot supply ordinary arbitrary-jump additions. This report does not contain a differential walk, sign/lift-safe relation labels, or a cost ratio to rho.
Machine data: `results/kummer-model-gate.jsonl` and `results/kummer-model-gate.replay.jsonl`.